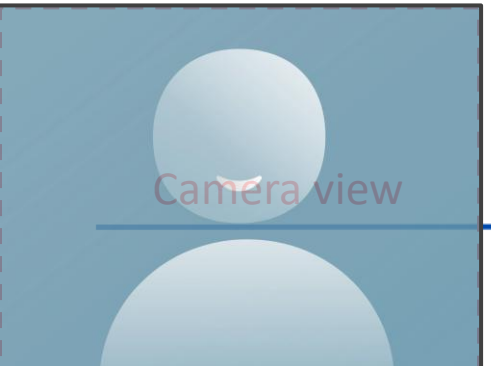
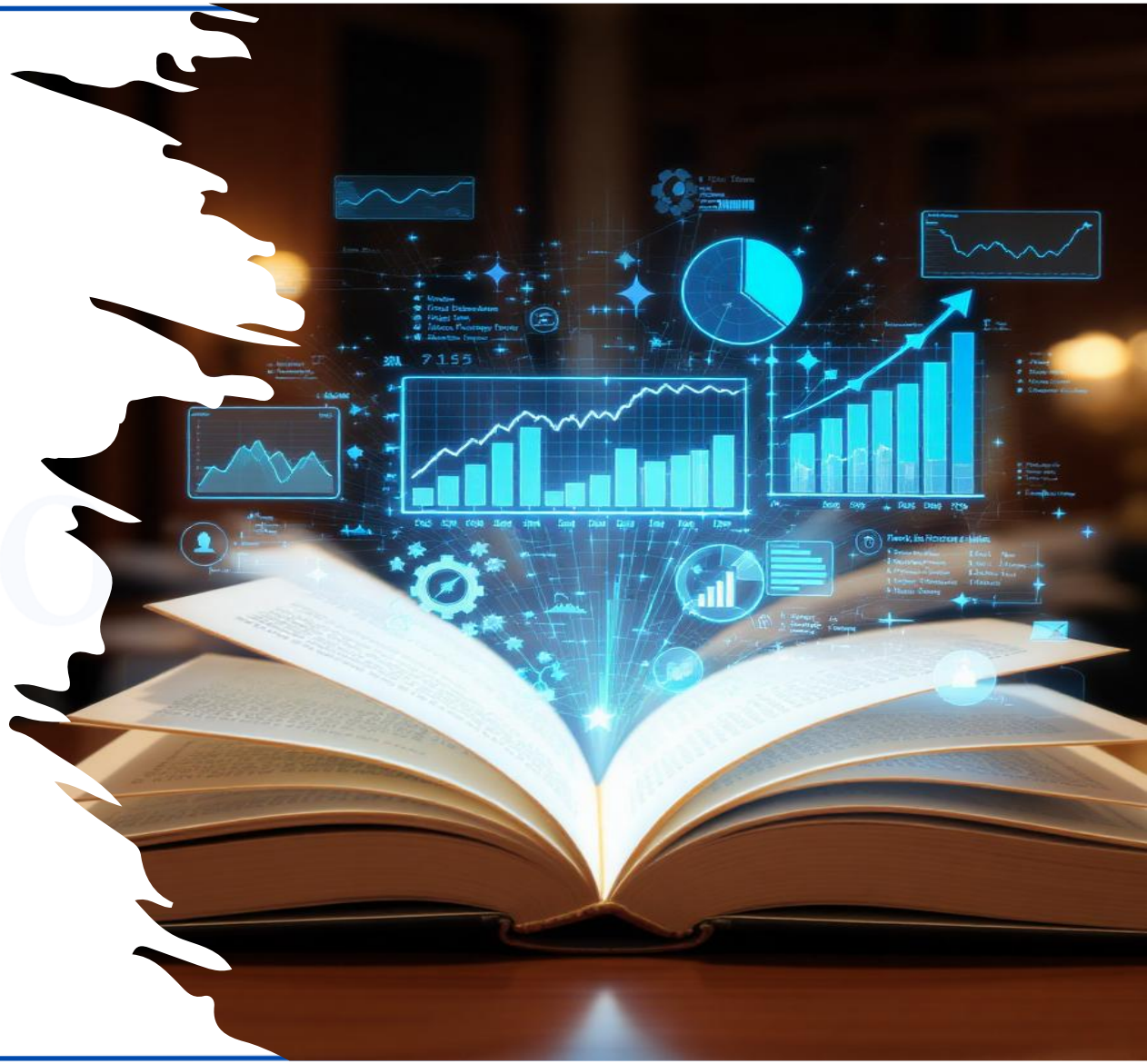
Optimizing Book Recommendation Systems through Integration of Sentiment Analysis on User Reviews

Rizqi Fajar, Putu Harry Gunawan, I Wayan Palton Anuwiksa

Telkom University





INTRODUCTION

The Challenge

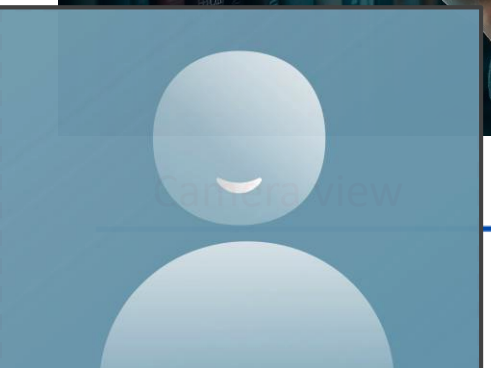
Exponential growth of books online makes finding matches difficult.

Current Limitations

Traditional systems rely on ratings but overlook emotional insights.

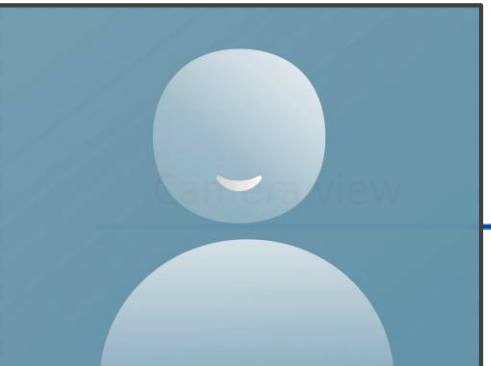
Our Approach

Integrating sentiment analysis with collaborative filtering improves recommendation accuracy.



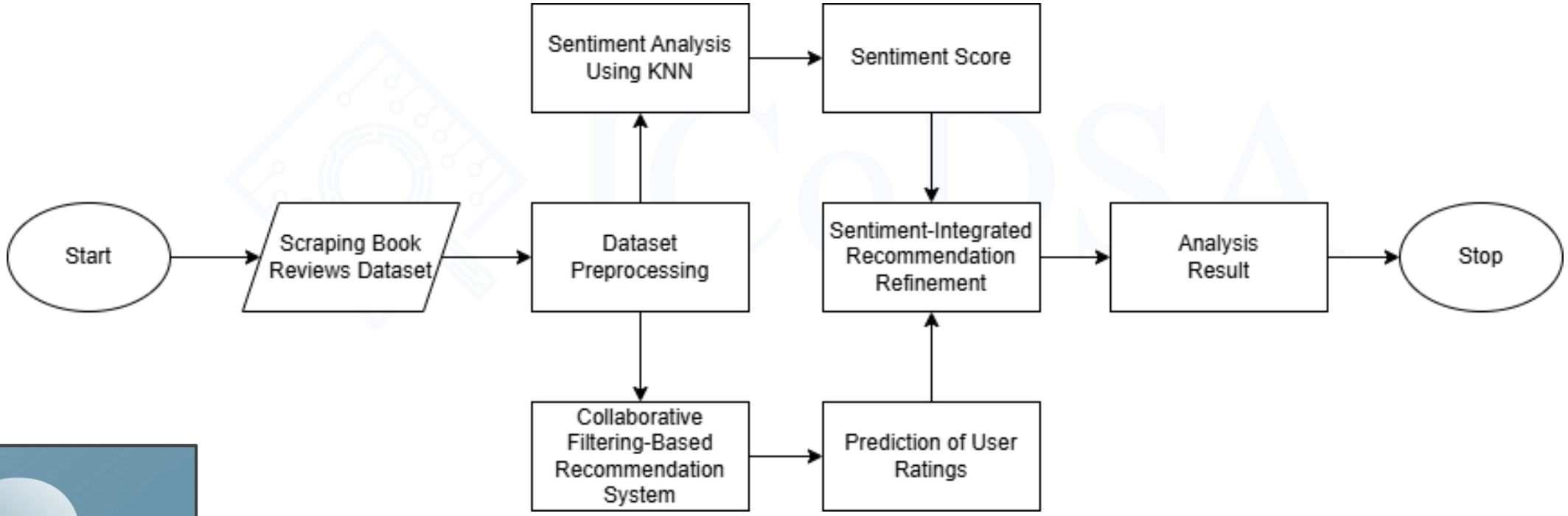
RESEARCH OBJECTIVES

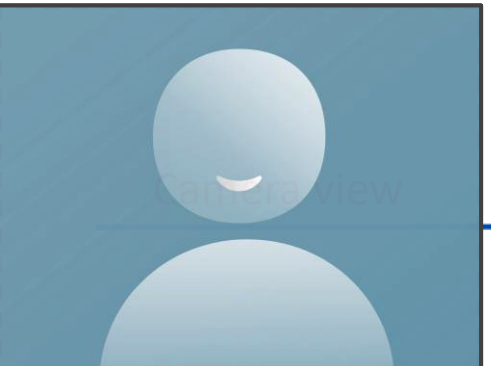
1. To perform sentiment analysis on user reviews using the K-Nearest Neighbors (KNN) algorithm to classify reviews as positive or negative.
2. To build a book recommendation system using a user-based collaborative filtering approach, recommending books based on user preferences.
3. To integrate sentiment analysis results into the recommendation system to enhance the quality of the recommendations, providing more relevant and personalized suggestions to users.



METHODOLOGY

System Design





METHODOLOGY

Dataset

Source

Goodreads reviews from a single author for consistency.

Volume

18,978 total reviews analyzed.

Key Attributes

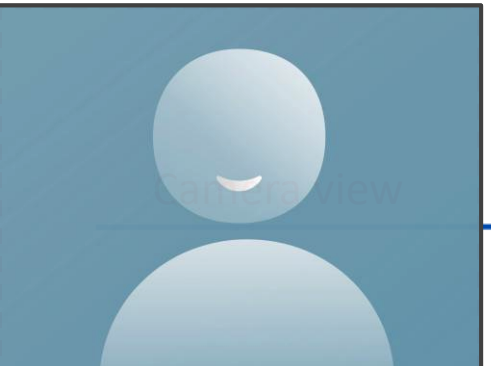
- Author name
- Book title
- Username
- Rating (1-5 scale)
- Review text

Processing

Removed irrelevant characters, noise, stopwords, tokenized, lemmatized, and normalized text.

TABLE I
SAMPLE OF SCRAPED DATASET

Author	Book Title	Username	Rating	Review
Carissa Broadbent	The Songbird & the Heart of Stone	User 1	Rating 5 out of 5	I AM THE CHOSEN ONE ALL MY LOVE to Carissa...
Carissa Broadbent	The Serpent and the Wings of Night	User 2	Rating 4 out of 5	sarah janet walked so this brutal shit could run...
Carissa Broadbent	The Ashes & the Star-Cursed King	User 3	Rating 3 out of 5	hmm. i feel like im missing something because everyone...
Carissa Broadbent	Slaying the Vampire Conqueror	User 4	Rating 2 out of 5	just like broadbent's second book, the ashes and the...
Carissa Broadbent	Children of Fallen Gods	User 5	Rating 1 out of 5	What in the internalized misogyny is this shit? I've...



METHODOLOGY

Sentiment Analysis

Sentiment Analysis with KNN:

- **KNN** with **TF-IDF** vectorization to classify sentiment:

1. Convert reviews to numerical features using TF-IDF.
2. Compute Euclidean distance:

$$d(x, y) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$

Where:

x_i and y_i = the features of the two reviews being compared.

n = the total number of features.

3. Classify reviews based on the majority sentiment of nearest neighbors.

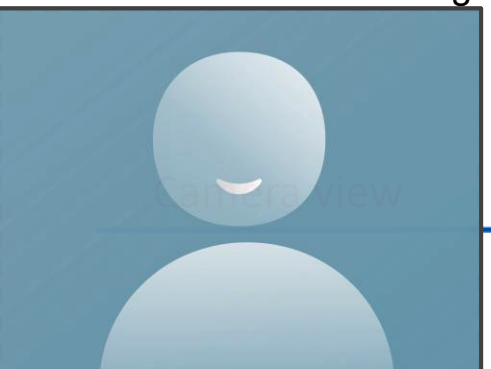
Sentiment Score Formula:

$$\text{Sentiment Score} = \frac{\sum_{i=1}^{N} \text{Sentiment}_i}{N}$$

Where:

Sentiment_i = 1 for positive reviews, 0 for negative reviews.

N = Total number of reviews.



METHODOLOGY

Recommendation System

User-Based Collaborative Filtering:

- **Cosine Similarity** to measure similarity between users' rating patterns.

Cosine Similarity Formula:

$$\text{cos-sim}(\mathbf{r}_u, \mathbf{r}_v) = \frac{\mathbf{r}_u \cdot \mathbf{r}_v}{\|\mathbf{r}_u\| \|\mathbf{r}_v\|}$$

Where:

r_u, r_v = Rating vectors for users u and v.

Predicted Rating Formula:

$$\hat{r}_{u,i} = \bar{r}_u + \frac{\sum_{v \in N(u)} \text{cos-sim}(\mathbf{r}_u, \mathbf{r}_v)\left(r_{v,i} - \bar{r}_v\right)}{\sum_{v \in N(u)} \left|\text{cos-sim}(\mathbf{r}_u, \mathbf{r}_v)\right|}$$

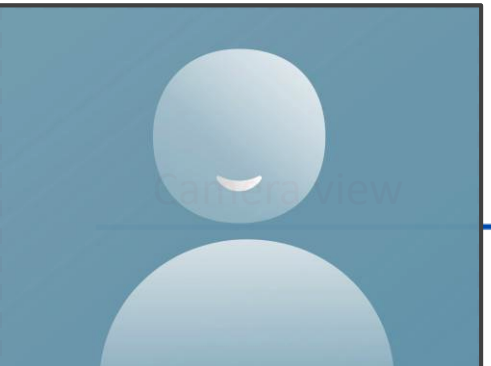
Where:

$\hat{r}_{u,i}$ = Predicted rating for user u on item i.

$\bar{r}_u, \bar{r}_v$ = Average ratings for users u and v.

$r_{v,i}$ = Rating given by user v to item i.

$N(u)$ = Set of neighbors of user u.



METHODOLOGY

Sentiment-Integrated Recommendation Refinement

Sentiment-Integrated Refinement:

- **Combine predicted ratings** with sentiment scores for final ranking.

Final Ranking Formula:

$$R_j = \alpha \cdot \hat{r}_{ij} + \beta \cdot S_j^{\text{normalized}}$$

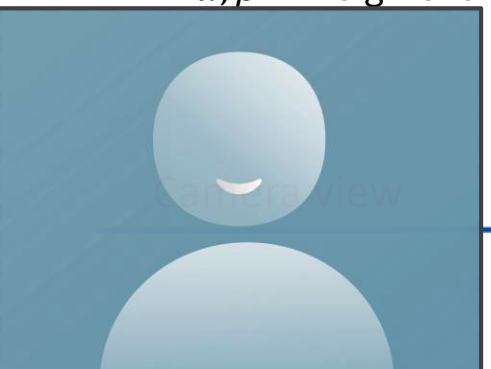
Where:

R_j = Final score for book j.

$\hat{r}_{ij}$ = Predicted rating for book j.

$S_{\text{normalized}_j}$ = Normalized sentiment score for book j.

α, β = Weights for balancing ratings and sentiment.



RESULT

TABLE II
SENTIMENT ANALYSIS PERFORMANCE METRICS

Sentiment	Precision	Recall	F1-Score
Positive	0.85	0.82	0.83
Negative	0.83	0.86	0.85

K-Nearest Neighbors Model Accuracy : 84 %

TABLE VI
PERFORMANCE COMPARISON: WITH AND WITHOUT SENTIMENT INTEGRATION

Model	RMSE	MAE
Without Sentiment Integration	1.3704	1.1894
With Sentiment Integration	1.3288	0.8381

Evaluation Result:

RMSE (Root Mean Squared Error):

Reduced from 1.3704 to 1.3288.

MAE (Mean Absolute Error):

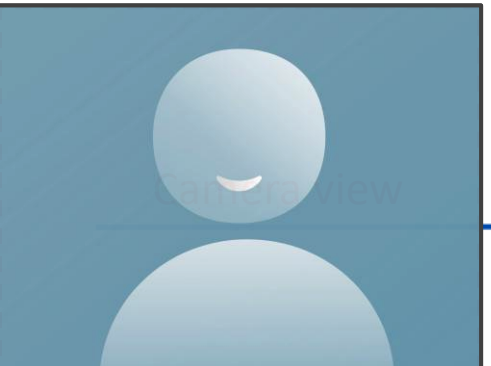
Reduced from 1.1894 to 0.8381.

TABLE III
RECOMMENDATIONS WITHOUT SENTIMENT INTEGRATION

Rank	Book Name	Predicted Rating
1	Daughter of No Worlds	5.0000
2	The Songbird & the Heart of Stone	2.2811
3	Realm of Darkness	2.0474
4	Flirting with Darkness	1.7006
5	Ashen Son	1.1514
6	Fierce Hearts	1.0000

TABLE V
RECOMMENDATIONS WITH SENTIMENT INTEGRATION

Book Name	Predicted Rating	Sentiment Score	Final Score
The Songbird & the Heart of Stone	2.2811	5.0000	12.2811
Daughter of No Worlds	5.0000	1.9024	8.8048
Fierce Hearts	1.0000	3.0129	7.0258
Realm of Darkness	2.0474	2.3978	6.8431
Flirting with Darkness	1.7006	1.0000	3.7006
Ashen Son	1.1514	1.1276	3.4067



CONCLUSION

Improved Accuracy

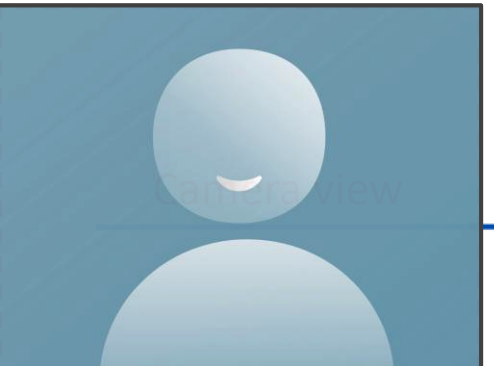
Reduced RMSE from 1.3704 to 1.3288 and MAE from 1.1894 to 0.8381.

Emotional Relevance

Prioritized books with high positive sentiment scores.

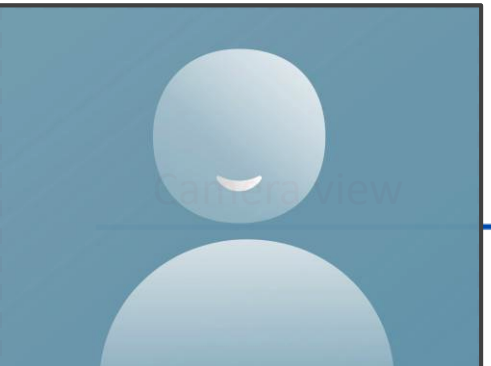
Limitations

Imbalanced dataset with fewer negative reviews affected results.



Future Work

Expand datasets and explore transformer-based architectures.

THANK YOU

rizqifajar@student.telkomuniversity.ac.id, phgunawan@telkomuniversity.ac.id, paltonanuwiksa@telkomuniversity.ac.id